

Tutorial 8: FINA2205

Time Series

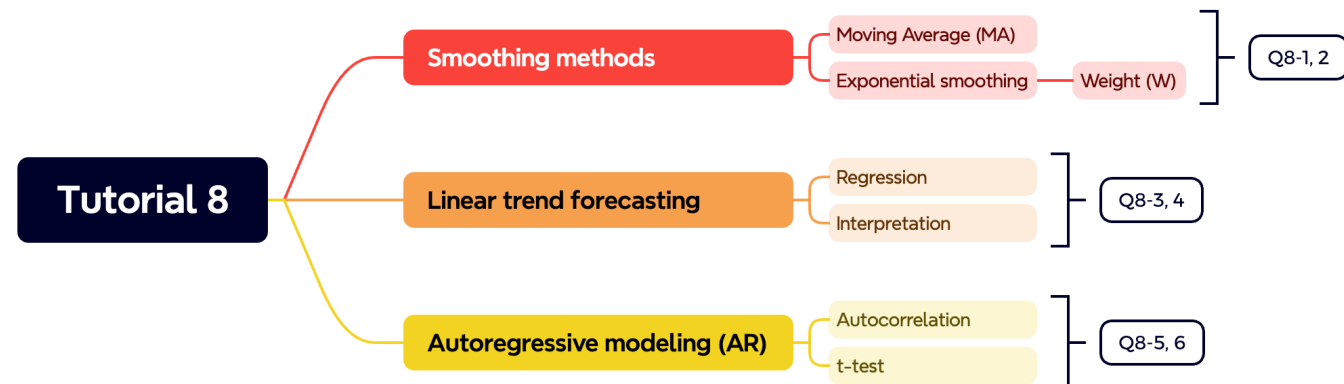
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May 2026

Agenda

- 6 problems



Recap: Smoothing methods

What is a Time Series?

- Definition: A sequence of data points that occur in successive order over some period of time.
- Data structure: Consists of numerical data obtained at regular time intervals (e.g., annual GDP and daily stock price).
- Visualizing: Plotted on a two-dimensional graph where the vertical axis represents the variable of interest, and the horizontal axis represents time.

Recap: Smoothing methods

The components of a Time Series

$$T = \textit{Trend} + \textit{Seasonal} + \textit{Cyclical} + \textit{Irregular}$$

1. Trend: The overall, persistent, long-term movement (upward or downward).
2. Seasonal: Regular, periodic, wave-like patterns usually within a 1-year period.
3. Cyclical: Repeating swings or wave-like patterns that occur over more than one year.
4. Irregular: Erratic, unpredictable, or residual fluctuations are often considered the “**noise**” in the data.

Recap: Smoothing methods

Why use Smoothing Methods?

- The problem: Raw time series is often filled with erratic “noise” and irregular components, making the underlying trend hard to spot.
- By “smoothing” these fluctuations, the general direction becomes **easier** to observe.
- Smoothing methods help answer the question: “*Is there a meaningful trend behind the fluctuations?*”

Recap: Smoothing methods

Smoothing Method 1 – Moving Averages (MA)

- Concept: A centered MA smooths a time series by replacing each value with the **average** of surrounding observations.

- For a 5-year centered moving average:

$$MA_t = \frac{Y_{t-2} + Y_{t-1} + Y_t + Y_{t+1} + Y_{t+2}}{5}$$

- The calculated average is assigned to the **middle year**.

Recap: Smoothing methods

Moving Averages

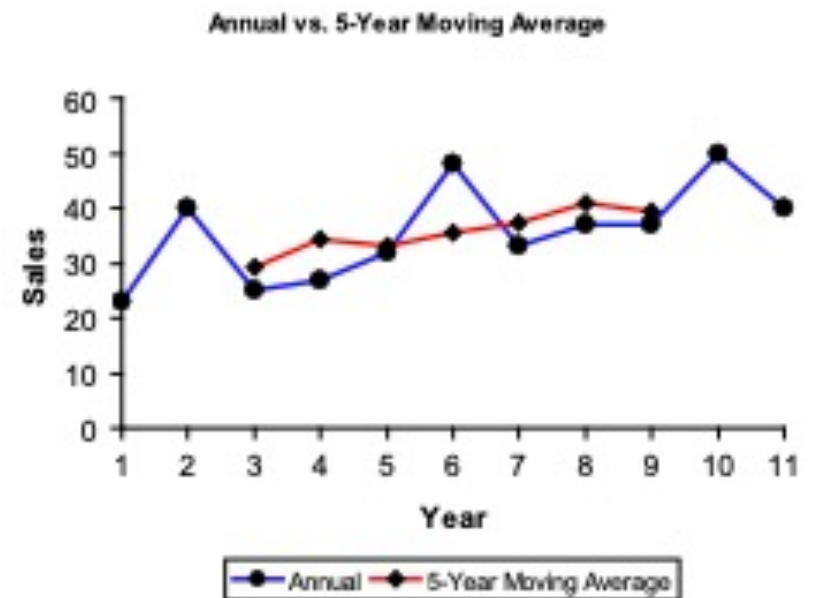
Year	Sales	Average Year	5-Year Moving Average
1	23	3	29.4
2	40	4	34.4
3	25	5	33.0
4	27	6	35.4
5	32	7	37.4
6	48	8	41.0
7	33	9	39.4
8	37	...	...
9	37		
10	50		
11	40		

etc...

- Each moving average is for a consecutive block of 5 years

$$3 = \frac{1+2+3+4+5}{5}$$

$$29.4 = \frac{23+40+25+27+32}{5}$$



Recap: Smoothing methods

Smoothing Method 2 – Exponential Smoothing

- Concept: A **weighted** moving average where the weights decline exponentially.
- The formula:

$$E_1 = Y_1$$
$$E_i = WY_i + (1 - W)E_{i-1}$$

can be used as the forecast for year $i+1$ ($\hat{Y}_{i+1}$)!

For $i = 2, 3, 4, \dots$

where:

E_i = exponentially smoothed value for period i

E_{i-1} = exponentially smoothed value already
computed for period $i - 1$

Y_i = observed value in period i

W = weight (smoothing coefficient), $0 < W < 1$

Recap: Smoothing methods

$$\begin{aligned}
 E_i &= WY_i + (1 - W)E_{i-1} \\
 &\rightarrow E_i = WY_i + (1 - W)(WY_{i-1} + (1 - W)E_{i-2}) \\
 &\rightarrow E_i = WY_i + (1 - W)WY_{i-1} + (1 - W)^2E_{i-2} \\
 &\rightarrow E_i = WY_i + (1 - W)WY_{i-1} + (1 - W)^2(WY_{i-2} + (1 - W)E_{i-3}) \\
 &\rightarrow E_i = WY_i + (1 - W)WY_{i-1} + (1 - W)^2WY_{i-2} + (1 - W)^3E_{i-3} \\
 &\rightarrow E_i = WY_i + (1 - W)WY_{i-1} + (1 - W)^2WY_{i-2} + \dots
 \end{aligned}$$

- Clearly, $W > (1 - W)W > (1 - W)^2W$ ($0 < W < 1$)
- Conclusion: This shows that recent observations receive **larger** weight, while older observations receive weights that decline exponentially.

Recap: Smoothing methods

- The parameter: W is the smoothing coefficient, **subjectively** chosen, and ranges from 0 to 1.
- Large W (close to 1): Provides less smoothing, tracking recent data closely.
- Small W (close to 0): Provides more smoothing, excellent for stripping away unwanted cyclical and irregular components to see the long-term trend.
- The rule: Use a **smaller** smoothing when we want to focus on the long-term pattern.

8-2

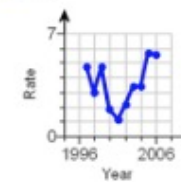
The data in the accompanying table represent the three-month interest rates for a certain investment from 1996 to 2005. Complete parts (a) through (f).

Year	Rate
1996	5.56
1997	5.70
1998	3.35
1999	3.40
2000	2.17
2001	1.13
2002	1.84
2003	4.71
2004	3.03
2005	4.76

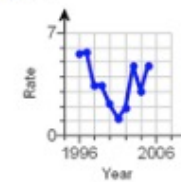
a. Plot the data.

Choose the correct graph below.

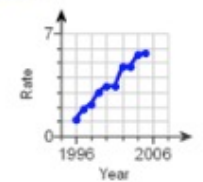
☐ A.



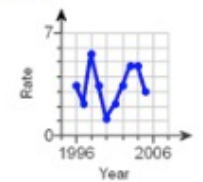
☐ C.



☐ B.



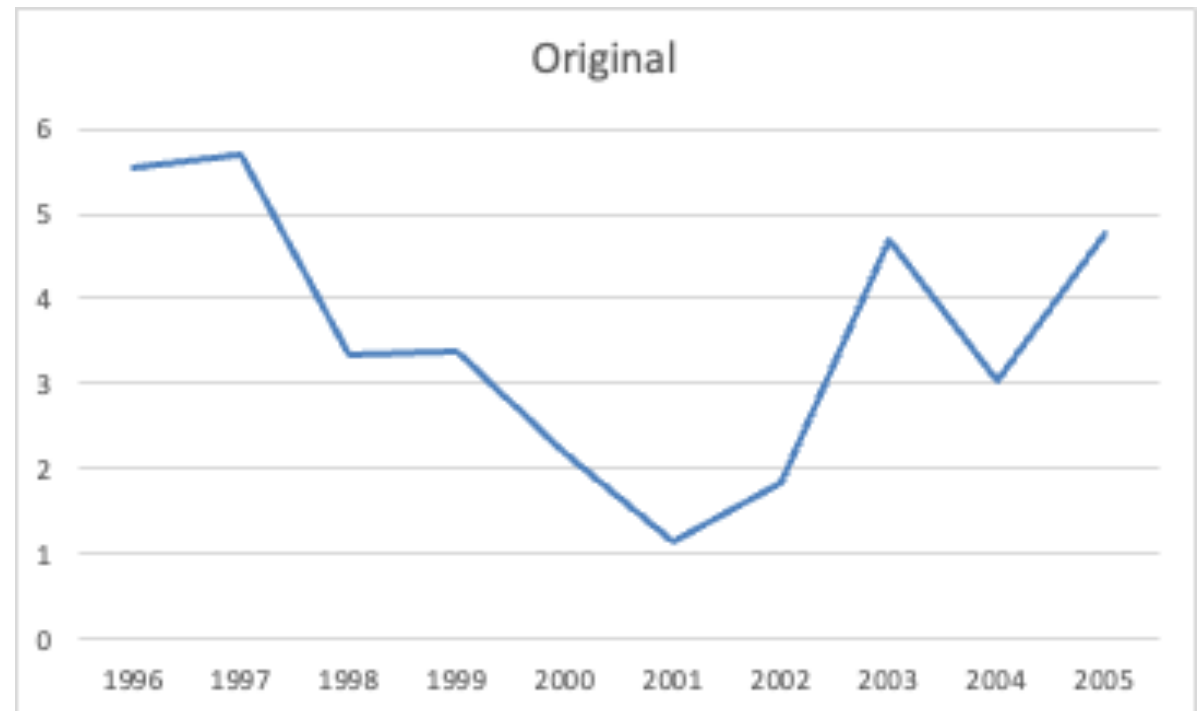
☐ D.



8-2 solution

- C

Year	Original
1996	5.56
1997	5.7
1998	3.35
1999	3.4
2000	2.17
2001	1.13
2002	1.84
2003	4.71
2004	3.03
2005	4.76



8-2 (cont.)

b. Fit a three-year moving average to the data and plot the results.

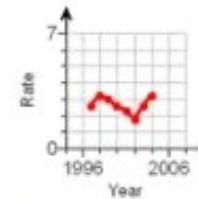
Complete the table below.

Year	Rate	MA 3-Year
1996	5.56	
1997	5.70	
1998	3.35	
1999	3.40	
2000	2.17	
2001	1.13	
2002	1.84	
2003	4.71	
2004	3.03	
2005	4.76	

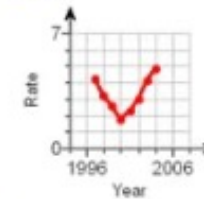
(Round to two decimal places as needed.)

Plot the three-year moving average. Choose the correct graph below.

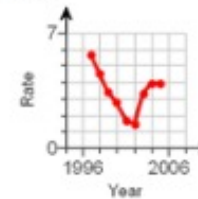
☐ A.



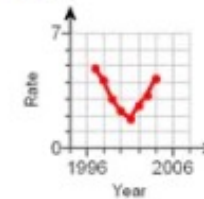
☐ B.



☐ C.



☐ D.



8-2 solution (cont.)

b.

First average (**Year 1997**)

$$\bullet MA(3) = \frac{5.56 + 5.7 + 3.35}{3} = 4.87$$

First average (Year 1998)

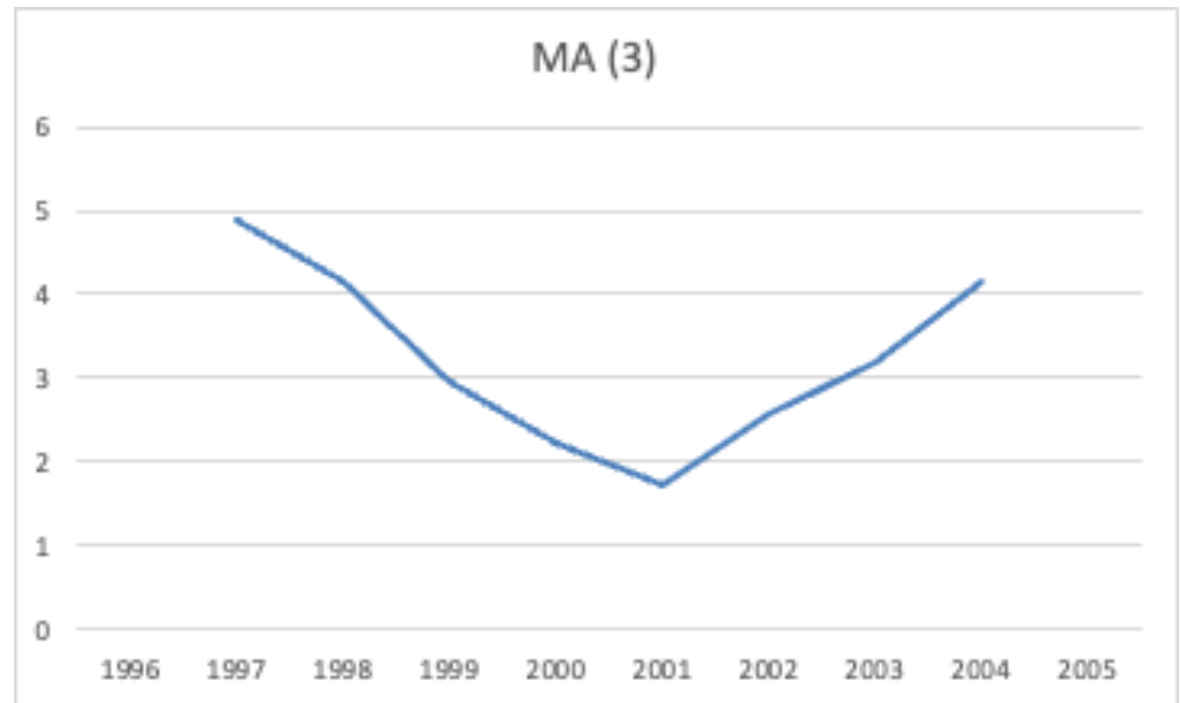
$$\bullet MA(3) = \frac{5.7 + 3.35 + 3.4}{3} = 4.15$$

• ...

Year	Original	MA (3)
1996	5.56	
1997	5.7	4.87
1998	3.35	4.15
1999	3.4	2.973333
2000	2.17	2.233333
2001	1.13	1.713333
2002	1.84	2.56
2003	4.71	3.193333
2004	3.03	4.166667
2005	4.76	

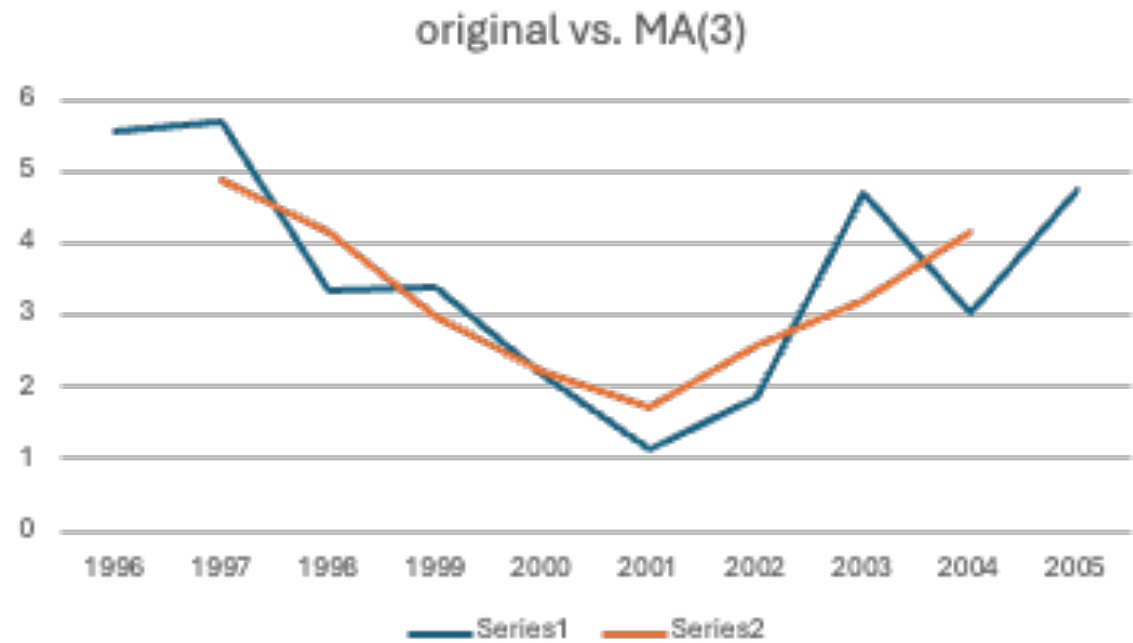
8-2 solution (cont.)

- D



8-2 solution (cont.)

- D



8-2 (cont.)

c. Using a smoothing coefficient of $W = 0.80$, exponentially smooth the series and plot the results.

Complete the table below.

Year	Rate	Exponentially Smoothed Series ($W = 0.80$)
1996	5.56	
1997	5.70	
1998	3.35	
1999	3.40	
2000	2.17	
2001	1.13	
2002	1.84	
2003	4.71	
2004	3.03	
2005	4.76	

(Round to two decimal places as needed.)

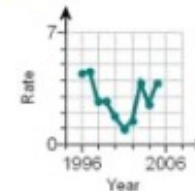
d. What is your exponentially smoothed forecast for 2006?

(Round to two decimal places as needed.)

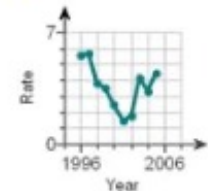
$$\Rightarrow \hat{Y}_{2006} = E_{2005} = ?$$

Plot the exponentially smoothed series. Choose the correct graph below.

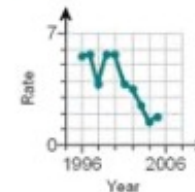
☐ A.



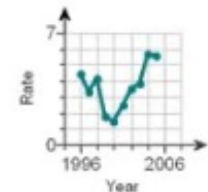
☐ B.



☐ C.



☐ D.



8-2 solution (cont.)

c.

The formula:

$$E_1 = Y_1$$
$$E_i = WY_i + (1 - W)E_{i-1}$$

- $E_{1996} = Y_1 = 5.56$
- $E_{1997} = WY_{1997} + (1 - W)E_{1996} = 0.8(5.7) + 0.2(5.56) = 5.672$
- $E_{1998} = WY_{1998} + (1 - W)E_{1997} = 0.8(3.35) + 0.2(5.672) = 3.8144$

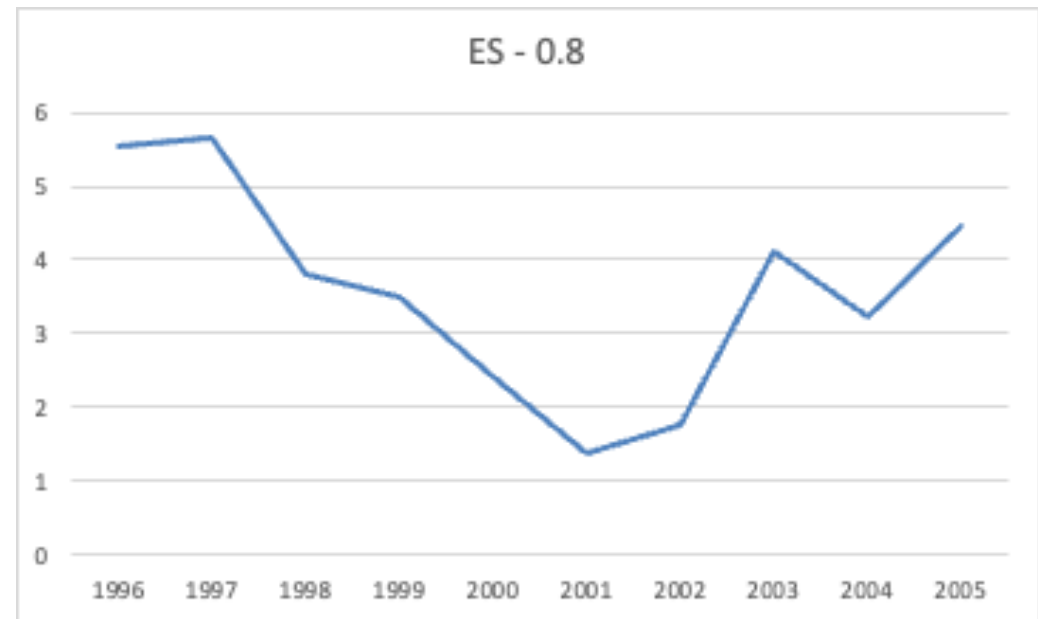
Year	Original	0.8
1996	5.56	5.56
1997	5.7	5.672
1998	3.35	3.8144
1999	3.4	3.48288
2000	2.17	2.432576
2001	1.13	1.390515
2002	1.84	1.750103
2003	4.71	4.118021
2004	3.03	3.247604
2005	4.76	4.457521

8-2 solution (cont.)

- **B**

d.

$$\begin{aligned}\hat{Y}_{2006} &= E_{2005} \\ &= WY_{2005} + (1 - W)E_{2004} \\ &= 0.8(4.76) + 0.2(3.247604) \\ &= 4.457521 \approx 4.46\end{aligned}$$



8-2 (cont.)

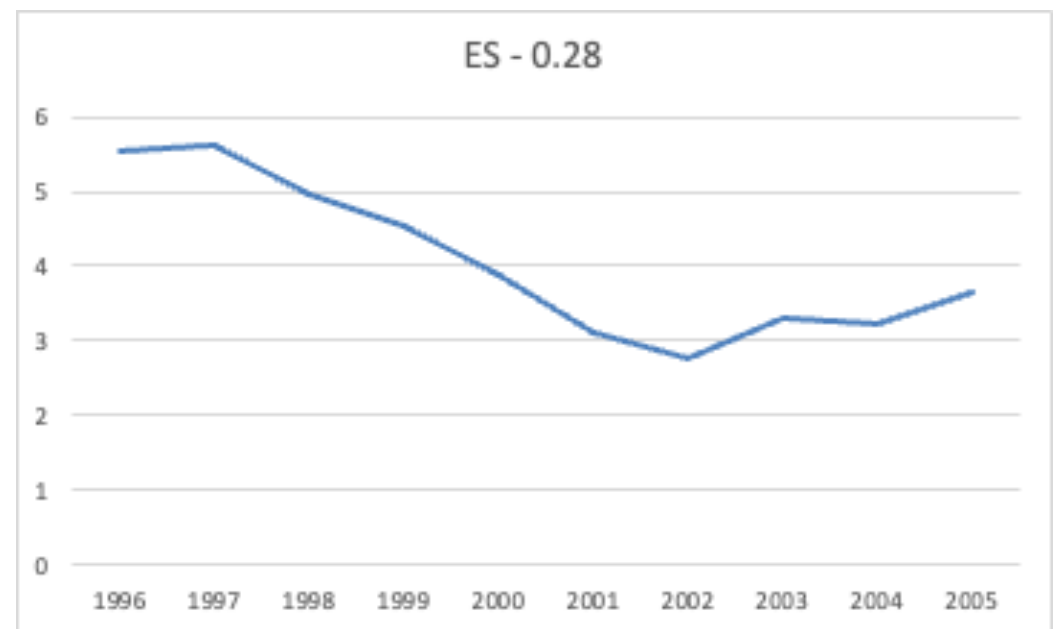
e. Repeat (c) and (d), using a smoothing coefficient of $W = 0.28$.

8-2 solution (cont.)

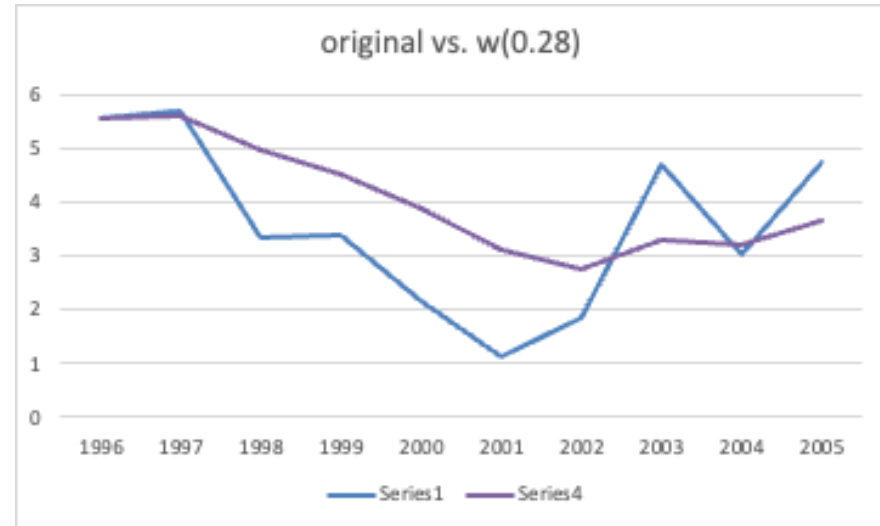
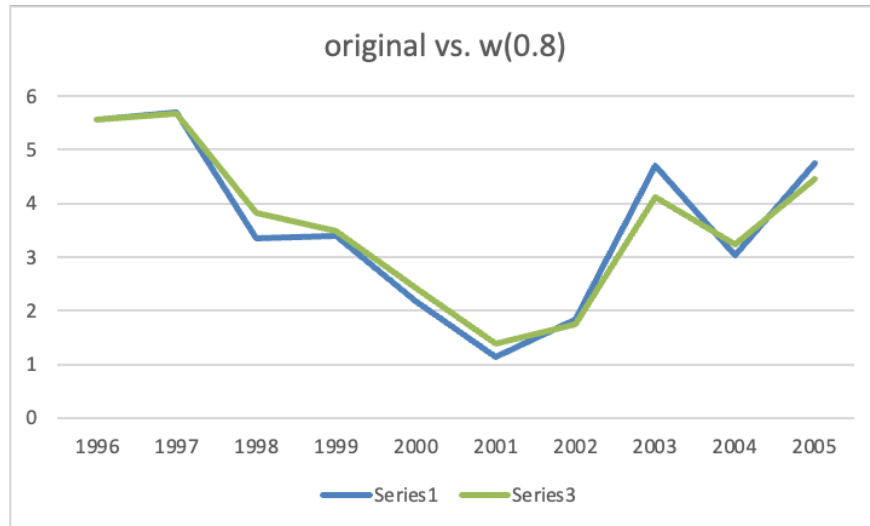
e.

$$\begin{aligned}\hat{Y}_{2006} &= E_{2005} \\ &= 3.653296 \\ &\approx 3.65\end{aligned}$$

Year	0.28
1996	5.56
1997	5.5992
1998	4.969424
1999	4.529985
2000	3.869189
2001	3.102216
2002	2.748796
2003	3.297933
2004	3.222912
2005	3.653296



8-2 solution (cont.)



8-1

1. You are using exponential smoothing on an annual time series concerning total revenues (in millions of dollars). You decide to use a smoothing coefficient of $W = 0.4$, and the exponentially smoothed value for 2010 is $E_{2010} = (0.4)(12.1) + (0.6)(9.1)$.
- a. What is the smoothed value of this series in 2010?
 - b. What is the smoothed value of this series in 2011 if the value of the series in that year is \$11.6 million?
-

a. $E_{2010} = \$$ _____ million

(Type an integer or a decimal.)

b. $E_{2011} = \$$ _____ million

(Type an integer or a decimal.)

8-1 solution

a.

The formula:

$$E_i = WY_i + (1 - W)E_{i-1}$$

- $E_{2010} = WY_{2010} + (1 - W)E_{2009} = 0.4(12.1) + 0.6(9.1) = 10.3$

b.

- As $W = 0.4$, $Y_{2011} = 11.6$, and $E_{2010} = 10.3$,

- $E_{2011} = WY_{2011} + (1 - W)E_{2010} = 0.4(11.6) + 0.6(10.3) = 10.82$

Recap: Linear trend forecasting

The limitation of smoothing

- Moving averages (MA) and exponential smoothing are descriptive tools used to **filter out** noise and identify the current trend.
- However, they are generally limited to **short-term** forecasting.

The power of regression

- Linear trend forecasting uses regression analysis to estimate a **persistent** path over time, helping us to make forecasts far into the future.

Recap: Linear trend forecasting

The core equation and parameters (Week 5 Tutorial 3)

- The formula

$$\hat{Y} = b_0 + b_1X$$

- $\hat{Y}$ (Dependent variable): The forecast or predicted value of the variable of interest.
- X (Independent variable): The time period.
- b_0 (Y-intercept): The starting point of our trend line.
- b_1 (Slope): The angle of our trend line, indicating the rate of change.

Recap: Linear trend forecasting

- The “coded year” trap: We should not plug **actual calendar years** (like “2007” or “2026”) directly into the X variable. This will break the model!
- Coded year (X): Time periods are numbered **starting with 0** for the base year, and increasing by 1 for each subsequent time period.

Rule of thumb for calculating X

1. Target year minus base year ($X = Y_{target} - Y_{base}$)
2. If a year is the n -th year in the dataset, its coded time value is

$$X = n_{target} - 1$$

Recap: Linear trend forecasting

- Example: If the dataset starts in 2007, then 2007 is the 1st year ($X = 0$), 2008 is the 2nd year ($X = 2 - 1 = 1$), and 2009 is the 3rd year ($X = 3 - 1 = 2$).

X !

Calendar year	Sequential year (n)	Coded year
2007	1	0
2008	2	1
2009	3	2
2010	4	3
2011	5	4
2012	6	5
2013	7	6

Recap: Linear trend forecasting

Interpreting the Y-intercept (b_0) (Week 5 Tutorial 3)

- Statistical meaning: The predicted value of Y when $X = 0$.
- It reflects the predicted baseline value at the very **beginning** of the time series data.

Interpreting the slope (b_1)

- Statistical meaning: **The average change** in $\hat{Y}$ for every 1-unit increase in X .
- It represents the predicted annual growth rate (if positive), or annual rate of decline (if negative).

8-3

3. The linear trend forecasting equation for an annual time series containing 37 values (from 1970 to 2006) on net sales (in billions of dollars) is shown below. Complete (a) through (e) below.

$$\hat{Y}_t = 4.9 + 0.4X_t$$

a. Interpret the Y-intercept, b_0 . Which statement below is the interpretation of the Y-intercept?

- ☐ A. The Y-intercept $b_0 = 0.4$ reflects the predicted net sales in 1970.
- ☐ B. The Y-intercept $b_0 = 0.4$ indicates that sales are predicted to increase by \$0.4 billion/year.
- ☐ C. The Y-intercept $b_0 = 4.9$ indicates that sales are predicted to increase by \$4.9 billion/year.
- ☐ D. The Y-intercept $b_0 = 4.9$ reflects the predicted net sales in 1970.

b. Interpret the slope, b_1 . Which statement below is the interpretation of the slope?

- ☐ A. The slope $b_1 = 4.9$ reflects the predicted net sales in 1970.
- ☐ B. The slope $b_1 = 0.4$ indicates that sales are predicted to increase by \$0.4 billion/year.
- ☐ C. The slope $b_1 = 4.9$ indicates that sales are predicted to increase by \$4.9 billion/year.
- ☐ D. The slope $b_1 = 0.4$ reflects the predicted net sales in 1970.

8-3 solution

a/b.

- **D** (The Y-intercept $b_0 = 4.9$ reflects the predicted net sales in 1970).
- **B** (The slope $b_1 = 0.4$ indicates that the sales are predicted to increase by \$0.4 billion per year).

8-3 (cont.)

c. What is the fitted trend value for the fifth year?

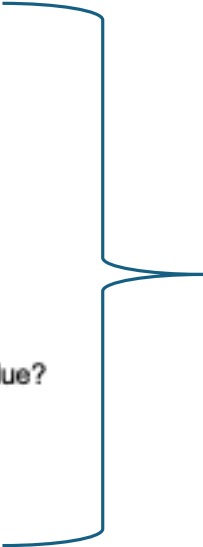
\$ _____ billion
(Simplify your answer.)

d. What is the fitted trend value for the most recent year?

\$ _____ billion
(Simplify your answer.)

e. What is the projected trend forecast two years after the last value?

\$ _____ billion
(Simplify your answer.)



Coded year? $\rightarrow X = n_{target} - 1$

8-3 solution (cont.)

c.

$$\hat{Y} = 4.9 + 0.4X$$

The rule: the coded value for any n -th year is $X = n_{target} - 1$. For the fifth year, $X = 5 - 1 = 4$

$$\hat{Y} = 4.9 + 0.4(4) = 6.5$$

d.

The linear trend forecasting equation for an annual time series containing 37 values (from 1970 to 2006)

- The series has 37 values, so the most recent is the 37th year ($n = 37$). The coded year is $X = 37 - 1 = 36$

$$\hat{Y} = 4.9 + 0.4(36) = 19.3$$

8-3 solution (cont.)

e.

- Take the last value's X (which is 36) and add 2. The coded year for two year after the last value is $X = 36 + 2 = 38$

$$\hat{Y} = 4.9 + 0.4(38) = 20.1$$

8-4

4. Gross domestic product (GDP) is a major indicator of a nation's overall economic activity. The GDP (in billions of current dollars) for a nation from 1987 to 2006 is shown in the following data table. Complete (a) through (d) below.

¹ Click the icon to view the data table.

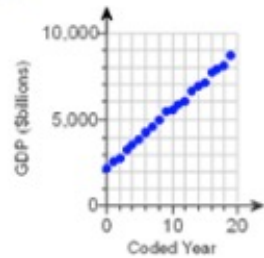
1: More Info

				Full data set
Year	GDP	Year	GDP	
1987	2,212.2	1997	5,579.6	
1988	2,593.3	1998	5,876.5	
1989	2,770.9	1999	6,060.1	
1990	3,289.7	2000	6,589.7	
1991	3,598.3	2001	6,895.0	
1992	3,881.0	2002	7,161.3	
1993	4,247.4	2003	7,742.0	
1994	4,522.0	2004	7,903.9	
1995	4,972.7	2005	8,147.7	
1996	5,422.0	2006	8,730.1	

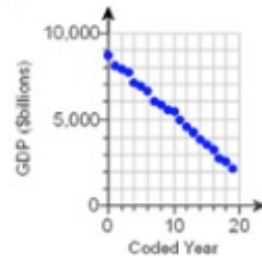
8-4

a. Plot the data. Choose the correct graph below.

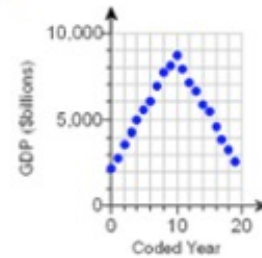
☐ A.



☐ B.



☐ C.



b. Compute a linear trend forecasting equation.

$$\hat{Y}_t = \underline{\hspace{2cm}} + (\underline{\hspace{2cm}}) X_t$$

(Round to two decimal places as needed.)

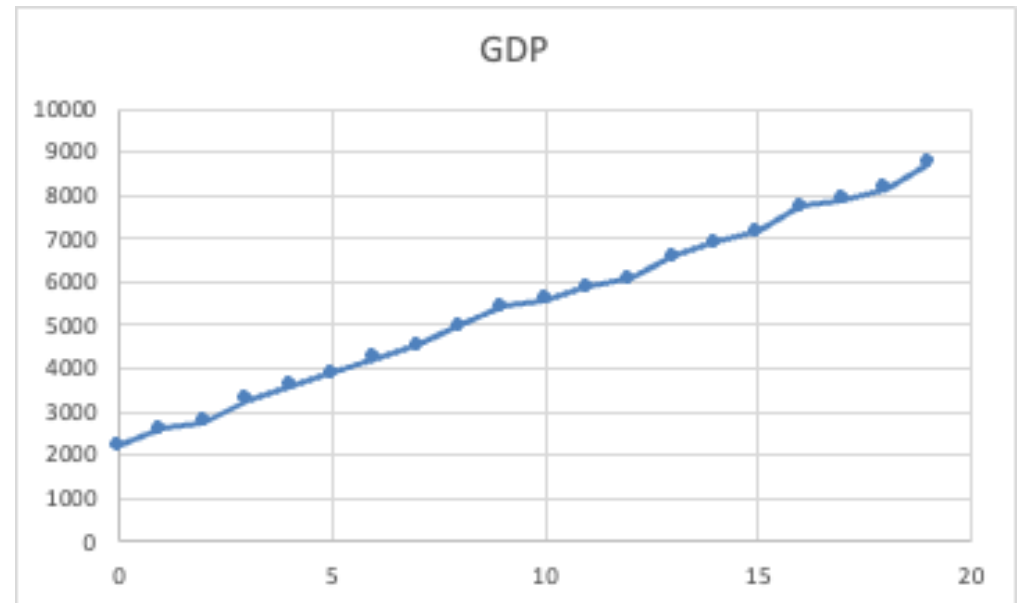
8-4 solution

Use the coded year as the explanatory variable!

a.

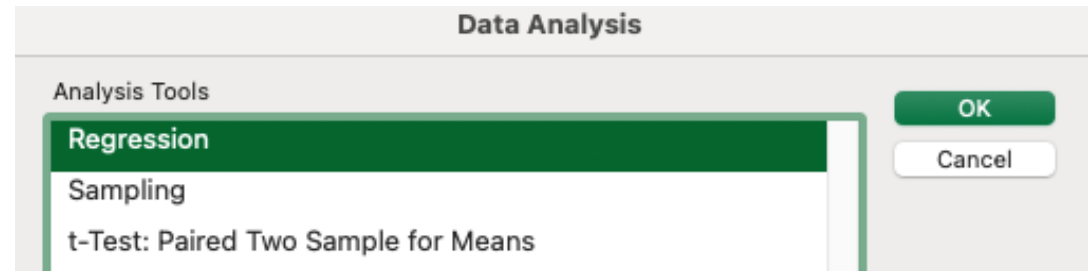
A

Q4	X	Y
1987	0	2212.2
1988	1	2593.3
1989	2	2770.9
1990	3	3289.7
1991	4	3598.3
1992	5	3881
1993	6	4247.4
1994	7	4522
1995	8	4972.7
1996	9	5422
1997	10	5579.6
1998	11	5876.5
1999	12	6060.1
2000	13	6589.7
2001	14	6895
2002	15	7161.3
2003	16	7742
2004	17	7903.9
2005	18	8147.7
2006	19	8730.1



8-4 solution

b.



SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.998926							
R Square	0.997853							
Adjusted R	0.997733							
Standard Error	94.64299							
Observations	20							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	74919480	74919480	8364.073	1.8E-25			
Residual	18	161231.3	8957.296					
Total	19	75080711						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	2221.096	40.786	54.45731	1.97E-21	2135.408	2306.784	2135.408	2306.784
X Variable 1	335.6499	3.670098	91.45531	1.8E-25	327.9393	343.3605	327.9393	343.3605

$$\hat{Y} = 2221.10 + 335.65X \text{ (round to two decimal places)}$$

8-4 (cont.)

c. What are your forecasts for 2007 and 2009?

The forecast for the GDP in 2007 is \$_____ billion.
(Round to one decimal place as needed.)

The forecast for the GDP in 2009 is \$_____ billion.
(Round to one decimal place as needed.)



Coded year? $\rightarrow X = Y_{target} - Y_{base}$

d. What conclusions can you reach concerning the trend in GDP?

There is (1) _____ trend in GDP since 1987.

- (1) ☐ an upward
☐ a downward
☐ no significant

8-4 solution (cont.)

c.

$$X = Y_{target} - Y_{base}$$
$$X_{2007} = 2007 - 1987 = 20; X_{2009} = 2009 - 1987 = 22;$$

- The equation: $\hat{Y} = 2221.10 + 335.65X$
- $\hat{Y}_{2007} = 2221.096 + 335.6499 (20) = 8934.09$
- $\hat{Y}_{2009} = 2221.096 + 335.6499 (22) = 9605.39$

d.

- **A** (Since the slope is **positive**, the model indicates an **upward** trend in GDP since 1987).

Q4	X	Y
1987	0	2212.2
1988	1	2593.3
1989	2	2770.9
1990	3	3289.7
1991	4	3598.3
1992	5	3881
1993	6	4247.4
1994	7	4522
1995	8	4972.7
1996	9	5422
1997	10	5579.6
1998	11	5876.5
1999	12	6060.1
2000	13	6589.7
2001	14	6895
2002	15	7161.3
2003	16	7742
2004	17	7903.9
2005	18	8147.7
2006	19	8730.1

Recap: Autoregressive modeling

Motivation for autoregressive modeling

- So far, we have known how to filter out noise (Smoothing Methods) and how to map long-term trajectories based on the passage of time (Linear Trends).
- However, these models assume that deviations from the trend are just random noise. They fail to capture the **immediate momentum** of the data.
- Autoregressive models are designed to capture that kind of internal dependence.

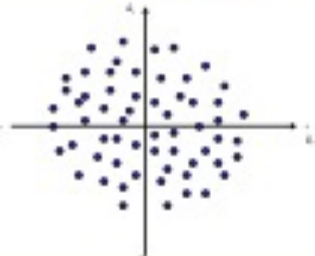
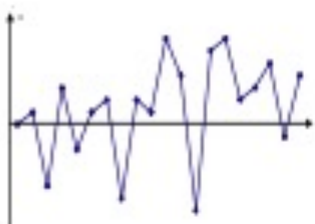
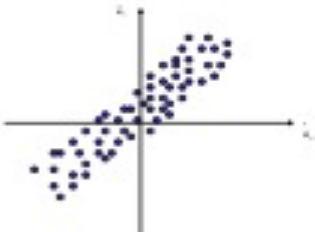
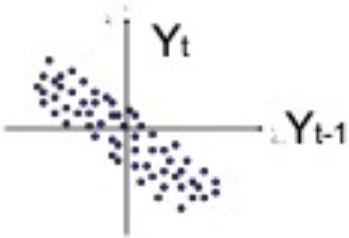
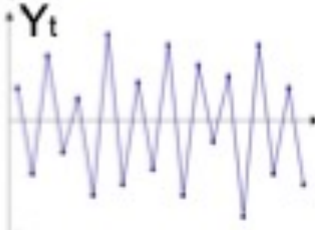
Recap: Autoregressive modeling

- Big question: *What if next year's GDP is driven less by the year number, and more by this year's GDP?*
- In business and finance, data often has a “**memory**.” A high value today may tend to be followed by a high value tomorrow.
- This internal relationship, where a variable is correlated with its own past values, is called **autocorrelation**.
- Instead of looking outside the variable for an explanation, we are going to look inside.

Recap: Autoregressive modeling

Autocorrelation: Order and direction

- Order means how many lags we look back.
- If Y_i is related to Y_{i-1} , we have *first-order* autocorrelation.
- If Y_i is related to Y_{i-2} , we have *second-order* autocorrelation.
- If Y_i is related to Y_{i-3} , we have *third-order* autocorrelation.
- Positive autocorrelation means high values tend to be followed by high values, and low values by low values.
- Negative autocorrelation means high values tend to be followed by low values, or vice versa.

Auto correlation	Scatter plot of Y_t against Y_{t-1}	Time Series Line plot of Y_t
No auto correlation		
Positive auto correlation		
Negative auto correlation		

Recap: Autoregressive modeling

- Autoregressive modeling (AR) takes advantage of this relationship for forecasting.
- p^{th} -order autoregressive model (AR(p) model)

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + \cdots + A_p Y_{i-p} + \delta_i$$

- AR(1): use one lag, Y_{i-1} .
- AR(2): use two lags, Y_{i-1} and Y_{i-2} .
- AR(3): use three lags, Y_{i-1} , Y_{i-2} and Y_{i-3} .
- The larger p is, the more past information we include.

Recap: Autoregressive modeling

How do we choose “ p ”?

- We use a **step-down approach**, and usually start with the **highest-order** AR model.
- Test whether the highest lag is actually useful.
- More lags do NOT automatically mean a better model. They can make the model unnecessarily complicated.
- The goal: Achieve model parsimony (informative but not overcomplicated).

Recap: Autoregressive modeling

- The AR(p) Model

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + \dots + A_p Y_{i-p} + \delta_i$$

- The AR(3) Model

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + A_3 Y_{i-3}$$

- The AR(2) Model

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2}$$

- The AR(1) Model

$$Y_i = A_0 + A_1 Y_{i-1}$$

If insignificant



Recap: Autoregressive modeling

T-test systematic approach (Week 6 Tutorial 4)

1. State the hypotheses.
2. Calculate the t statistic.
3. Find the critical values.
4. Compare and conclude.

Recap: Autoregressive modeling

Step 1: State the hypotheses.

- For an AR(3) model, we test whether the third lag should be included:

$$H_0: A_3 = 0$$

$$H_1: A_3 \neq 0$$

- Note, by convention, the null hypothesis always states “no effect”. And the alternative hypothesis is what we are trying to find evidence for (e.g., the third lag contributes to the model).

Recap: Autoregressive modeling

Step 2: Calculate the t statistic.

$$t = \frac{a_3 - 0}{S_{a_3}}$$

where a_3 is the estimated coefficient for Y_{i-3} , S_{a_3} is its standard error, and 0 is the hypothesized value under H_0 .

- In practice, the estimated coefficient and standard error can be obtained from Excel's regression output.
- In the final exam, these values may be provided because the calculation is complex.

Recap: Autoregressive modeling

Step 3: Find the critical values.

- In an $AR(p)$ model, the sample size is $n-p$ rather than n !
- This is because the first p observations do not have enough lagged values.
- For example, in an $AR(2)$ model, the first *two* observations are lost because we need both the first and second lag.

Year	Y_i	Y_{i-1}	Y_{i-2}
05	4	--	--
06	3	4	--
07	2	3	4
08	3	2	3
09	2	3	2
10	2	2	3
11	4	2	2
12	6	4	2

Recap: Autoregressive modeling

- In an AR(p) model, the effective sample size is $n-p$.
- We estimate p lag coefficients plus 1 intercept.
- Therefore, the degrees of freedom are:

$$\begin{aligned} df &= (n - p) - p - 1 \\ &\rightarrow df = n - 2p - 1 \end{aligned}$$

For example, in an AR(3) model:

- $df = n - 2(3) - 1 = n - 7$

Recap: Autoregressive modeling

- After finding the degrees of freedom, we choose the significance level, usually:

$$\alpha = 0.05$$

- Because this is a **two-tailed test**, the rejection region is split into two tails:

$$\alpha/2 = 0.025$$

- We then use the t-table to find the value at:

$$t_{\alpha/2, df}$$

- Since the t-distribution is **symmetric**, the *two* critical values are:

$$+t_{\alpha/2, df} \text{ and } -t_{\alpha/2, df}$$

Recap: Autoregressive modeling

Step 4: Compare and conclude.

- If we reject H_0 :

$$t > +t_{\alpha/2,df} \text{ or } t < -t_{\alpha/2,df}$$

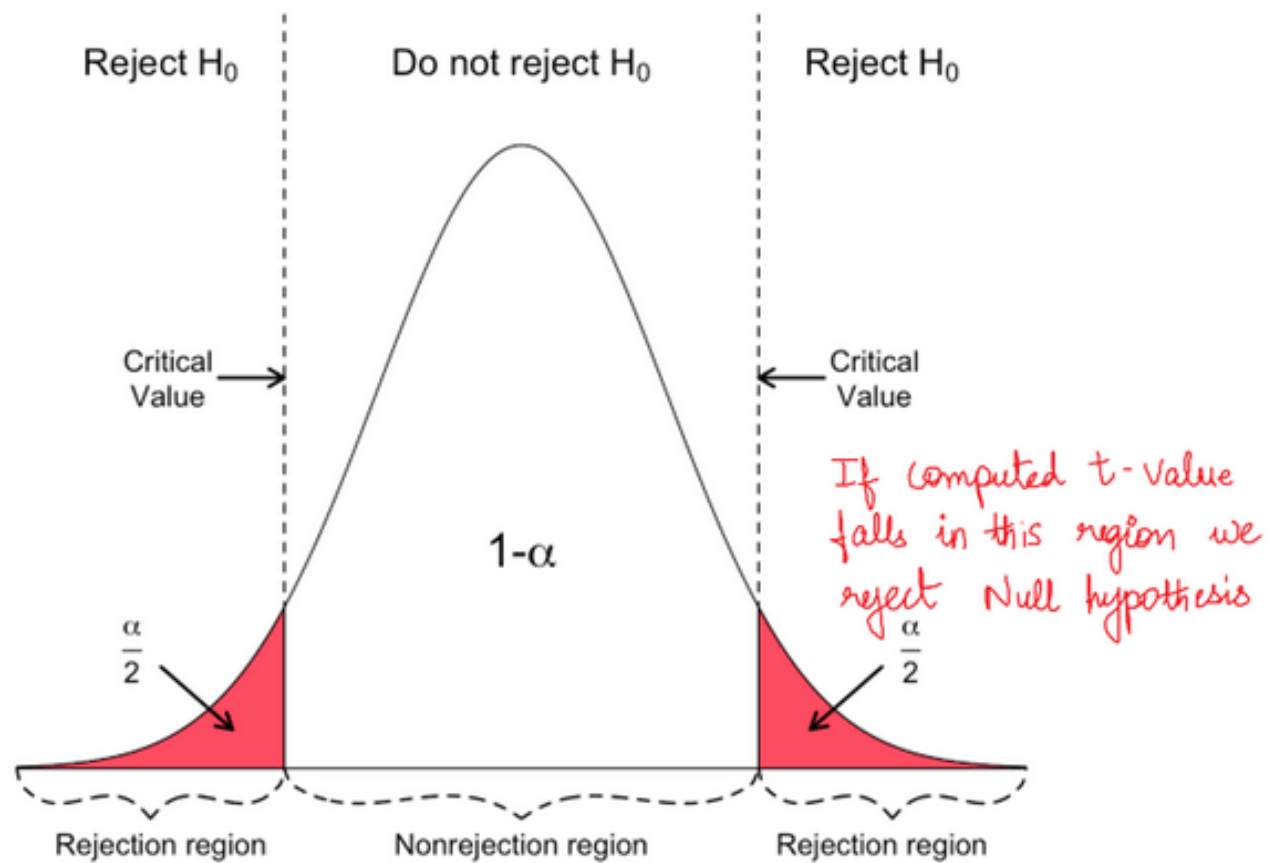
the third lag is statistically significant, so we keep the AR(3) model.

- If we do not reject H_0 :

$$-t_{\alpha/2,df} < t < +t_{\alpha/2,df}$$

the third lag is not statistically significant, so we move down to AR(2).

- Then we repeat the same logic for A_2 , and if necessary, for A_1 .



8-5

5. A third-order autoregressive model is fitted to an annual time series with 17 values and has the estimated parameters and standard errors shown below. At the 0.05 level of significance, test the appropriateness of the fitted model.

$$\begin{array}{cccc} a_0 = 4.63 & a_1 = 1.45 & a_2 = 0.87 & a_3 = 0.34 \\ S_{a_1} = 0.55 & S_{a_2} = 0.24 & S_{a_3} = 0.19 & \end{array}$$

² Click the icon to view the table for the critical values of t.

What are the hypotheses for this test?

- ☐ A. $H_0: A_3 \neq 0$
 $H_1: A_3 = 0$
- ☐ B. $H_0: A_2 = 0$
 $H_1: A_2 \neq 0$
- ☐ C. $H_0: A_3 = 0$
 $H_1: A_3 \neq 0$
- ☐ D. $H_0: A_2 \neq 0$
 $H_1: A_2 = 0$

8-5 solution

- Logic: We **always** test the significance of the **highest-order** parameter (A_p) in the model to see if the specific lag is useful. Here, $p = 3$.

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + \cdots + A_p Y_{i-p} + \delta_i$$

- *Step 1: State the hypotheses.*
- Null hypothesis (H_0): $A_3 = 0$ (the 3rd-order lag has no effect).
- Alternative hypothesis (H_1): $A_3 \neq 0$ (the 3rd-order lag is significant)
- **C** ($H_0: A_3 = 0, H_1: A_3 \neq 0$).

8-5 (cont.)

What is the test statistic for this test?

_____ (Round to four decimal places as needed.)

What are the critical values for this test?

(Round to four decimal places as needed. Use a comma to separate answers as needed.)

What is the result of the test of the appropriateness of the fitted model?

(1) _____ the null hypothesis. There is (2) _____ evidence to conclude that the third-order regression parameter is significantly different from zero, which means that the third-order autoregressive model (3) _____ appropriate.

- (1) ☐ Reject (2) ☐ sufficient (3) ☐ is
☐ Do not reject ☐ insufficient ☐ is not

8-5 solution (cont.)

Step 2: Calculate the t statistic.

$$t = \frac{a_3 - 0}{S_{a_3}}$$

$a_0 = 4.63$	$a_1 = 1.45$	$a_2 = 0.87$	$a_3 = 0.34$
	$S_{a_1} = 0.55$	$S_{a_2} = 0.24$	$S_{a_3} = 0.19$

- $t = \frac{0.34}{0.19} = 1.7895$

8-5 solution (cont.)

$$p = 3$$

$$n = 17$$

$$\alpha = 0.05$$

A third-order autoregressive model is fitted to an annual time series with 17 values and has the estimated parameters and standard errors shown below. At the 0.05 level of significance, test the appropriateness of the fitted model.

$$\begin{array}{llll} a_0 = 4.63 & a_1 = 1.45 & a_2 = 0.87 & a_3 = 0.34 \\ S_{a_1} = 0.55 & S_{a_2} = 0.24 & S_{a_3} = 0.19 & \end{array}$$

$$\text{Degree of freedom (df)} = n - 2p - 1 = 10!$$

Step 3: Find the critical values.

- Lookup: Using a t-table for $df = 13$ and $\alpha/2 = 0.025$
- Critical values ($\pm t_{\alpha/2, df}$): ± 2.2281

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467

8-5 solution (cont.)

Step 4: Compare and conclude.

- Comparison: Our calculated t-stat (1.7895) is less than the upper critical value (2.2281).
- Decision: it falls in the “**Do not reject**” region.
- Final conclusion:

(1) _____ the null hypothesis. There is (2) _____ evidence to conclude that the third-order regression parameter is significantly different from zero, which means that the third-order autoregressive model (3) _____ appropriate.

(1) ☐ Reject
☒ Do not reject

(2) ☐ sufficient
☒ insufficient

(3) ☐ is
☒ is not

8-6

6. The accompanying data table shows the value, in dollars, of a certain stock index as an annual time series. Use the data to complete parts (a) through (d).

3: Stock Index Data Table

				Full data set	
Year	Value (\$)	Year	Value (\$)	Year	Value (\$)
1	1,334.00	11	3,165.30	20	9,093.60
2	635.00	12	3,040.70	21	11,100.30
3	1,136.20	13	3,149.90	22	10,369.10
4	1,423.40	14	3,244.70	23	10,190.90
5	1,512.20	15	3,702.10	24	7,961.60
6	1,449.60	16	4,160.40	25	10,671.50
7	1,243.90	17	5,241.00	26	10,788.00
8	1,640.70	18	6,670.70	27	10,717.50
9	1,703.10	19	8,107.80	28	12,463.20
10	2,110.10				

a. Fit a third-order autoregressive model to the stock index and test for the significance of the third-order autoregressive parameter. (Use $\alpha = 0.05$.)

What are the hypotheses for this test?

- ☐ A. $H_0: A_3 = 0$
 $H_1: A_3 \neq 0$
- ☐ B. $H_0: A_3 \neq 0$
 $H_1: A_3 = 0$
- ☐ C. $H_0: A_2 \neq 0$
 $H_1: A_2 = 0$
- ☐ D. $H_0: A_2 = 0$
 $H_1: A_2 \neq 0$

8-6

What is the test statistic for this test?

_____ (Round to four decimal places as needed.)

$\alpha = ? , s_{\alpha} = ?$
→ *Excel's Data Analysis tool*

What are the critical values for this test?

(Round to four decimal places as needed. Use a comma to separate answers as needed.)

What is the result of the test for the significance of the third-order autoregressive parameter?

- ☐ A. Reject H_0 . The third-order parameter of the autoregressive model is not significant and should not be included in the model.
- ☐ B. Do not reject H_0 . The third-order parameter of the autoregressive model is not significant and should not be included in the model.
- ☐ C. Do not reject H_0 . The third-order parameter of the autoregressive model is significant and should be included in the model.
- ☐ D. Reject H_0 . The third-order parameter of the autoregressive model is significant and should be included in the model.

8-6 solution

a.

Step 1: State the hypotheses.

- **A** ($H_0: A_3 = 0, H_1: A_3 \neq 0$).

Regression

Input

Input Y Range:

Input X Range:

☐ Labels

☐ Constant is Zero

☐ Confidence Level:

95

%

OK

Cancel

Q6	t	t-1	t-2	t-3
1	1334			
2	635	1334		
3	1136.2	635	1334	
4	1423.4	1136.2	635	1334
5	1512.2	1423.4	1136.2	635
6	1449.6	1512.2	1423.4	1136.2
7	1243.9	1449.6	1512.2	1423.4
8	1640.7	1243.9	1449.6	1512.2
9	1703.1	1640.7	1243.9	1449.6
10	2110.1	1703.1	1640.7	1243.9
11	3165.3	2110.1	1703.1	1640.7
12	3040.7	3165.3	2110.1	1703.1
13	3149.9	3040.7	3165.3	2110.1
14	3244.7	3149.9	3040.7	3165.3
15	3702.1	3244.7	3149.9	3040.7
16	4160.4	3702.1	3244.7	3149.9
17	5241	4160.4	3702.1	3244.7
18	6670.7	5241	4160.4	3702.1
19	8107.8	6670.7	5241	4160.4
20	9093.6	8107.8	6670.7	5241
21	11100.3	9093.6	8107.8	6670.7
22	10369.1	11100.3	9093.6	8107.8
23	10190.9	10369.1	11100.3	9093.6
24	7961.6	10190.9	10369.1	11100.3
25	10671.5	7961.6	10190.9	10369.1
26	10788	10671.5	7961.6	10190.9
27	10717.5	10788	10671.5	7961.6
28	12463.2	10717.5	10788	10671.5

Y range

X range

8-6 solution

Step 2: Calculate the *t* statistic.

- T-stat: $t = \frac{\alpha_3 - 0}{S_{\alpha_3}} = \frac{-0.0954}{0.2284} = -0.4176$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.9690517							
R Square	0.9390612							
Adjusted R Squar	0.9303557							
Standard Error	1032.6511							
Observations	25							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	345085482.7	1.15E+08	107.8694	6.41E-13			
Residual	21	22393733	1066368					
Total	24	367479215.7						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	432.65333	365.760911	1.182886	0.250074	-327.988	1193.295	-327.988	1193.295
X Variable 1	0.8650356	0.225043588	3.843858	0.000943	0.397032	1.333039	0.397032	1.333039
X Variable 2	0.2382129	0.295069856	0.80731	0.428538	-0.37542	0.851844	-0.37542	0.851844
X Variable 3	-0.0953987	0.228435141	-0.41762	0.680465	-0.57046	0.379658	-0.57046	0.379658

→ α_3 and S_{α_3} → t_3

8-6 solution

Step 3: Find the critical values.

- $df = (n - p) - p - 1 = (28 - 3) - 3 - 1 = 21$
- $\alpha = 0.05$
- From the t-table
- *Critical values* $(\pm t_{\alpha/2, df}) = \pm 2.0796$

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 solution

Step 4: Compare and conclude.

- Since $-2.0796 < t = -0.4176 < 2.0796$
- **B** (we do not reject the H_0 : The *third*-order parameter of the AR model is **not** significant).

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
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15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 (cont.)

b. If necessary, fit a second-order autoregressive model to the stock index and test for the significance of the second-order autoregressive parameter. (Use $\alpha = 0.05$.)

What are the hypotheses for this test?

- ☐ A. $H_0: A_1 \neq 0$
 $H_1: A_1 = 0$
- ☐ B. $H_0: A_2 = 0$
 $H_1: A_2 \neq 0$
- ☐ C. $H_0: A_2 \neq 0$
 $H_1: A_2 = 0$
- ☐ D. $H_0: A_1 = 0$
 $H_1: A_1 \neq 0$
- ☐ E. It is not necessary to fit a second-order model in this case.

What is the test statistic for this test? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. _____ (Round to four decimal places as needed.)
- ☐ B. It is not necessary to fit a second-order model in this case.

What are the critical values for this test? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. _____
(Round to four decimal places as needed. Use a comma to separate answers as needed.)
- ☐ B. It is not necessary to fit a second-order model in this case.

8-6 (cont.)

What is the result of the test for the significance of the second-order autoregressive parameter?

- ☐ A. Reject H_0 . The second-order parameter of the autoregressive model is not significant and should not be included in the model.
- ☐ B. Do not reject H_0 . The second-order parameter of the autoregressive model is not significant and should not be included in the model.
- ☐ C. Do not reject H_0 . The second-order parameter of the autoregressive model is significant and should be included in the model.
- ☐ D. Reject H_0 . The second-order parameter of the autoregressive model is significant and should be included in the model.
- ☐ E. It is not necessary to fit a second-order model in this case.

8-6 solution (cont.)

b.

- Since the third lag is not statistically significant, we move down to AR(2). And repeat the same logic for A_3 .

Step 1: State the hypotheses.

- **B** ($H_0: A_2 = 0, H_1: A_2 \neq 0$).

Q6	t	t-1	t-2
1	1334		
2	635	1334	
3	1136.2	635	1334
4	1423.4	1136.2	635
5	1512.2	1423.4	1136.2
6	1449.6	1512.2	1423.4
7	1243.9	1449.6	1512.2
8	1640.7	1243.9	1449.6
9	1703.1	1640.7	1243.9
10	2110.1	1703.1	1640.7
11	3165.3	2110.1	1703.1
12	3040.7	3165.3	2110.1
13	3149.9	3040.7	3165.3
14	3244.7	3149.9	3040.7
15	3702.1	3244.7	3149.9
16	4160.4	3702.1	3244.7
17	5241	4160.4	3702.1
18	6670.7	5241	4160.4
19	8107.8	6670.7	5241
20	9093.6	8107.8	6670.7
21	11100.3	9093.6	8107.8
22	10369.1	11100.3	9093.6
23	10190.9	10369.1	11100.3
24	7961.6	10190.9	10369.1
25	10671.5	7961.6	10190.9
26	10788	10671.5	7961.6
27	10717.5	10788	10671.5
28	12463.2	10717.5	10788

Y range

X range

8-6 solution (cont.)

Step 2: Calculate the *t* statistic.

$$\bullet \text{ T-stat: } t = \frac{\alpha_2}{S_{\alpha_2}} = \frac{0.1575}{0.2146} = 0.7338$$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.9704904							
R Square	0.9418516							
Adjusted R Square	0.9367953							
Standard Error	990.88917							
Observations	26							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	365782632.5	1.83E+08	186.27	6.2E-15			
Residual	23	22582811.13	981861.4					
Total	25	388365443.6						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	433.60082	332.3092264	1.304811	0.204862	-253.833	1121.035	-253.833	1121.035
X Variable 1	0.8576927	0.208650163	4.110674	0.000427	0.426067	1.289318	0.426067	1.289318
X Variable 2	0.1574453	0.214551822	0.733833	0.470465	-0.28639	0.60128	-0.28639	0.60128

→ α_2 and S_{α_2} → t_2

8-6 solution (cont.)

Step 3: Find the critical values.

- $df = (n - p) - p - 1 = (28 - 2) - 2 - 1 = 23$
- $\alpha = 0.05$
- From the t-table
- *Critical values* $(\pm t_{\alpha/2, df}) = \pm 2.0687$

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 solution (cont.)

Step 4: Compare and conclude.

- Since $-2.0687 < t = 0.7338 < 2.0687$
- **B** (we do not reject the H_0 : The second-order parameter of the AR model is **not** significant).

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 (cont.)

c. If necessary, fit a first-order autoregressive model to the stock index and test for the significance of the first-order autoregressive parameter. (Use $\alpha = 0.05$.)

What are the hypotheses for this test?

- ☐ A. $H_0: A_0 \neq 0$
 $H_1: A_0 = 0$
- ☐ B. $H_0: A_1 \neq 0$
 $H_1: A_1 = 0$
- ☐ C. $H_0: A_0 = 0$
 $H_1: A_0 \neq 0$
- ☐ D. $H_0: A_1 = 0$
 $H_1: A_1 \neq 0$
- ☐ E. It is not necessary to fit a first-order model in this case.

What is the test statistic for this test? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. _____ (Round to four decimal places as needed.)

~~☐ B. It is not necessary to fit a first-order model in this case.~~

What are the critical values for this test? Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. _____
(Round to four decimal places as needed. Use a comma to separate answers as needed.)

~~☐ B. It is not necessary to fit a first-order model in this case.~~

What is the result of the test for the significance of the first-order autoregressive parameter?

- ☐ A. Reject the null hypothesis. The first-order parameter of the autoregressive model is not significant and should not be included in the model.
- ☐ B. Do not reject the null hypothesis. The first-order parameter of the autoregressive model is not

Typo. C. Reject the null hypothesis. The first-order parameter of the autoregressive model is significant and should be included in the model

8-6 solution (cont.)

c.

- Since the second lag is still not statistically significant, we move down to AR(1). And repeat the same logic for A_2 and A_3 .

Step 1: State the hypotheses.

- **D** ($H_0: A_1 = 0, H_1: A_1 \neq 0$).

Q6	t	t-1
1	1334	
2	635	1334
3	1136.2	635
4	1423.4	1136.2
5	1512.2	1423.4
6	1449.6	1512.2
7	1243.9	1449.6
8	1640.7	1243.9
9	1703.1	1640.7
10	2110.1	1703.1
11	3165.3	2110.1
12	3040.7	3165.3
13	3149.9	3040.7
14	3244.7	3149.9
15	3702.1	3244.7
16	4160.4	3702.1
17	5241	4160.4
18	6670.7	5241
19	8107.8	6670.7
20	9093.6	8107.8
21	11100.3	9093.6
22	10369.1	11100.3
23	10190.9	10369.1
24	7961.6	10190.9
25	10671.5	7961.6
26	10788	10671.5
27	10717.5	10788
28	12463.2	10717.5

Y range

X range

8-6 solution (cont.)

Step 2: Calculate the *t* statistic.

- T-stat: $t = \frac{\alpha_1}{S_{\alpha_1}} = \frac{1.017}{0.0509} = 19.9799$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.970085							
R Square	0.9410649							
Adjusted R Square	0.9387074							
Standard Error	985.84631							
Observations	27							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	387974893	3.88E+08	399.1951	6.93E-17			
Residual	25	24297323.84	971893					
Total	26	412272216.8						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	326.50788	318.3073318	1.025763	0.314827	-329.058	982.0741	-329.058	982.0741
X Variable 1	1.0170657	0.05090453	19.97987	6.93E-17	0.912226	1.121906	0.912226	1.121906

→ α_1 and S_{α_1} → t_1

8-6 solution (cont.)

Step 3: Find the critical values.

- $df = (n - p) - p - 1 = (28 - 1) - 1 - 1 = 25$
- $\alpha = 0.05$
- From the t-table
- *Critical values* $(\pm t_{\alpha/2, df}) = \pm 2.0595$

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 solution (cont.)

Step 4: Compare and conclude.

- Since $t = 19.9799 > 2.0595$
- **C** (we reject the H_0 : The *first-order* parameter of the AR model is significant).

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	2.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874

8-6 (cont.)

d. If appropriate, forecast the value of the stock index for year 29. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

☐ A. \$ _____ (Round to the nearest cent as needed.)

☒ B. ~~Making a forecast is not appropriate.~~

8-6 solution (cont.)

d.
$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + \dots + A_p Y_{i-p} + \delta_i$$

- AR(1) model

$$Y_i = A_0 + A_1 Y_{i-1} + \sigma$$

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	326.50788	318.3073318	1.025763	0.314827	-329.058	982.0741	-329.058	982.0741
X Variable 1	1.0170657	0.05090453	19.97987	6.93E-17	0.912226	1.121906	0.912226	1.121906

$$\begin{aligned} \Rightarrow \hat{Y}_i &= 326.5079 + 1.0171 Y_{i-1} \\ \Rightarrow \hat{Y}_{29} &= 326.5079 + 1.0171(12463.2) \\ \Rightarrow \hat{Y}_{29} &= 13002.4 \end{aligned}$$

Q6	t	t-1
1	1334	
2	635	1334
3	1136.2	635
4	1423.4	1136.2
5	1512.2	1423.4
6	1449.6	1512.2
7	1243.9	1449.6
8	1640.7	1243.9
9	1703.1	1640.7
10	2110.1	1703.1
11	3165.3	2110.1
12	3040.7	3165.3
13	3149.9	3040.7
14	3244.7	3149.9
15	3702.1	3244.7
16	4160.4	3702.1
17	5241	4160.4
18	6670.7	5241
19	8107.8	6670.7
20	9093.6	8107.8
21	11100.3	9093.6
22	10369.1	11100.3
23	10190.9	10369.1
24	7961.6	10190.9
25	10671.5	7961.6
26	10788	10671.5
27	10717.5	10788
28	12463.2	10717.5

